SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – EXPERIMENTATION

Course Code:	ITA03	**Course Title:**	Mobile computing
CO Assessed:	CO2	**Assessment:**	Assessment Tool 2 – EXPERIMENTATION
Weightage:	30 % of CIA	**Total Marks:**	100 marks
CO2 Statement:	*Compare mobile communication technologies, including multiple access techniques and network evolution, based on performance and applications.*		
Student Name:	**Galla Sai Siddartha**	**Reg. No.:**	**192425093**
Section / Batch:	**2024 - AIML**	**Date:**	**24 - 10 - 2026**

Code of Conduct

I **Galla Sai Siddartha (192425093)** *(Name / Reg. No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: **Galla Sai Siddartha**

Instructions

- This test contains FIVE questions.
- Duration: 60 minutes (homework)

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

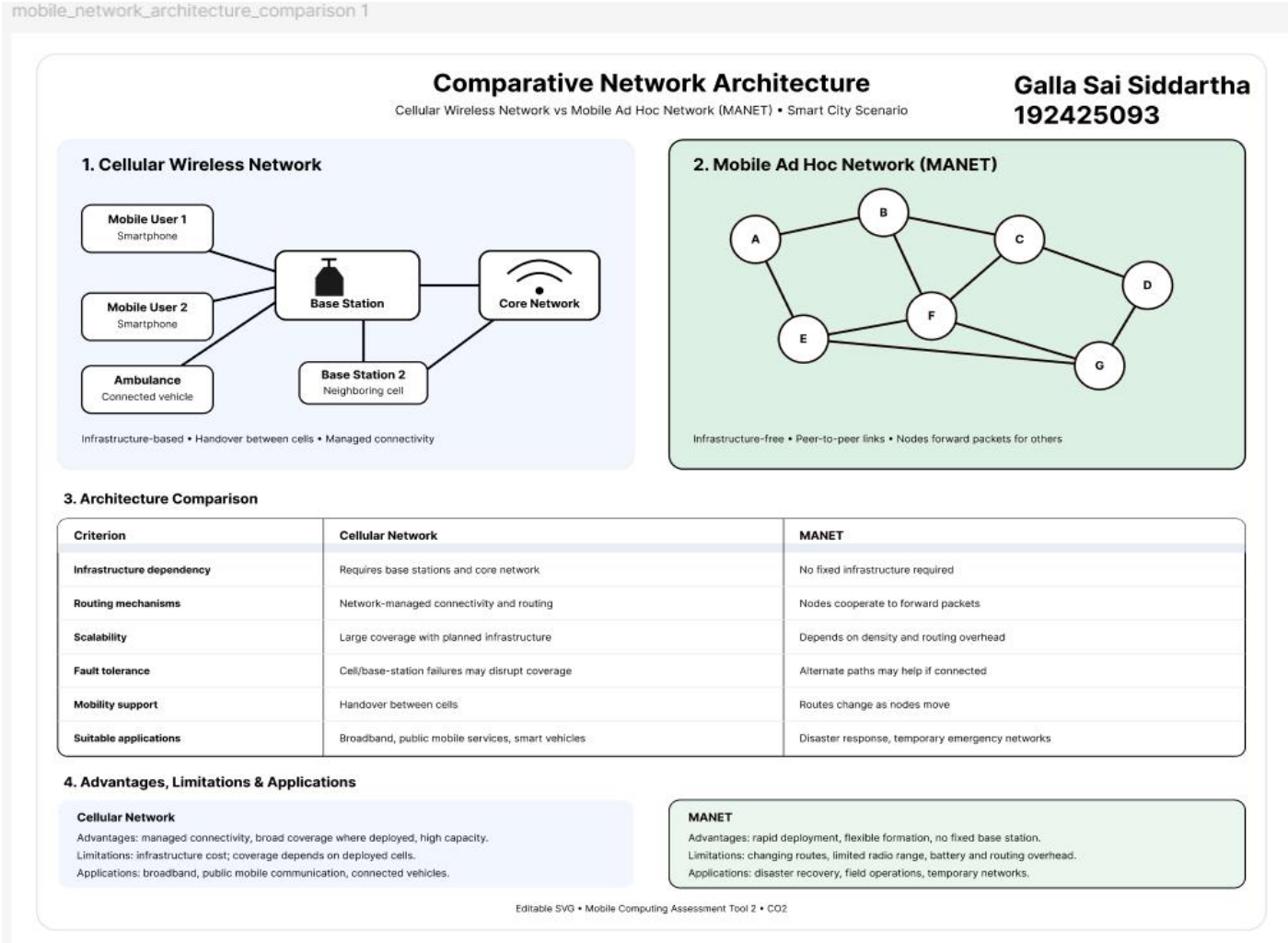
Annexure A - QUESTIONS

2. Comparative Network Architecture Design Using Figma (BL4 – Analyze)
Using Figma or draw.io, create a comparative architectural diagram showing both a Cellular Wireless Network and a Mobile Ad Hoc Network (MANET) operating in the same smart city environment.
Analyze the architectures by comparing:

- Infrastructure dependency.
- Routing mechanisms.
- Scalability.
- Fault tolerance.
- Mobility support.
- Suitable application scenarios.

Highlight the advantages and limitations of each architecture through annotations in your design.
Deliverables:

- Editable design file.
- GitHub repository with screenshots, comparison report, and design assets.



4. Mobile IP Application Design for Smart Healthcare (BL4 – Analyze)
Design a Smart Healthcare Mobile IP system using Figma that enables doctors, ambulances, hospitals, and patients to remain connected while moving across different wireless networks. Analyze:

- How Mobile IP maintains uninterrupted communication.
- Challenges during handoff between networks.
- Security vulnerabilities during mobility.
- Possible design improvements to increase reliability.

Provide navigation between screens demonstrating different mobility scenarios.
Deliverables:

- Interactive prototype.
- GitHub repository containing design files, exported prototype, and documentation.

